

Inequalities

(Some people collect butterflies, some collect stamps, and some collect Inequalities, R.P. Boas)

Basic Ingredients

1. For any real number x , $x^2 \geq 0$, and the equality holds only when $x = 0$. From this, we have $\sum_{i=1}^n x_i^2 \geq 0$, with equality holds only when $x_1 = x_2 = \cdots = x_n = 0$.
2. For any triangle with sides a , b and c , then $a+b-c > 0$, $a+c-b > 0$ and $b+c-a > 0$ (triangle inequality).

Important Inequalities

AM-GM-HM Inequality: For $a_1, a_2, \dots, a_n > 0$, then

$$AM = \frac{a_1 + a_2 + \cdots + a_n}{n} \geq GM = \sqrt[n]{a_1 a_2 \cdots a_n} \geq HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}},$$

and the equality holds if and only if $a_1 = a_2 = \cdots = a_n$.

Example: For $x > 0$, $x + \frac{1}{x} \geq 2\sqrt{x \cdot \frac{1}{x}} = 2$ with equality if and only if $x = 1$.

Example: For $a_1, a_2, \dots, a_n > 0$, by the AM-HM inequality, have

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}, \text{ hence } (a_1 + a_2 + \cdots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \right) \geq n^2.$$

Example: If $a, b, c > 0$, and $abc = 1$, find the minimum of $(a+b+c)(ab+bc+ca)$.

In fact $(a+b+c)(ab+bc+ca) \geq 3\sqrt[3]{abc} \cdot 3\sqrt[3]{(ab)(bc)(ca)} = 9$. The equality holds when $a = b = c = 1$.

Example: For a positive integer n , $(1 + \frac{1}{n+1})^{n+1} > (1 + \frac{1}{n})^n$.

Let $a_1 = a_2 = \cdots = a_n = 1 + \frac{1}{n}$ and $a_{n+1} = 1$. By the AM-GM inequality

$$\frac{n(1 + \frac{1}{n}) + 1}{n+1} = 1 + \frac{1}{n+1} > \sqrt[n+1]{(1 + \frac{1}{n})^n \cdot 1}. \text{ Taking the } n+1^{\text{th}} \text{ root to get the desired result.}$$

Example: Let a , b and c be the sides of a triangle, prove that $a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc$.

Let $x = b+c-a$, $y = c+a-b$ and $z = a+b-c$, then $x, y, z > 0$ (triangle inequality).

Solve to get $a = \frac{z+x}{2}$, $b = \frac{x+y}{2}$ and $c = \frac{y+z}{2}$. The inequality then becomes

$(\frac{z+x}{2})^2 y + (\frac{x+y}{2})^2 z + (\frac{y+z}{2})^2 x \leq \frac{3}{8}(z+x)(x+y)(y+z)$. This is equivalent to $x^2 y + y^2 z + z^2 x + xy^2 + yz^2 + zx^2 \geq 6xyz$, which follows from the AM-GM inequality.

Cauchy-Schwarz Inequality: For real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$,

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2),$$

equality holds if and only if $a_i b_j = a_j b_i$ for $1 \leq i, j \leq n$.

Example: Find the maximum of $a \cos \theta + b \sin \theta$, with $0 \leq \theta < 2\pi$.

In fact by the Cauchy-Schwarz inequality

$$(a \cos \theta + b \sin \theta)^2 \leq (a^2 + b^2)(\sin^2 \theta + \cos^2 \theta) = a^2 + b^2.$$

Hence $-\sqrt{a^2 + b^2} \leq a \cos \theta + b \sin \theta \leq \sqrt{a^2 + b^2}$. Equality holds when $a \cos \theta = b \sin \theta$.

Example: For real numbers a, b, c, d and e , $a + b + c + d + e = 8$ and

$$a^2 + b^2 + c^2 + d^2 + e^2 = 16. \text{ Find the maximum value of } e.$$

By the Cauchy-Schwarz $(a + b + c + d)^2 \leq (1^2 + 1^2 + 1^2 + 1^2)(a^2 + b^2 + c^2 + d^2)$, get

$$(8 - e)^2 \leq 4(16 - e^2). \text{ Solve to get } e(5e - 16) \leq 0. \text{ This implies } 0 \leq e \leq \frac{16}{5}. \text{ Examine the}$$

equality case of the Cauchy-Schwarz inequality to get $a = b = c = d = \frac{6}{5}$ and $e = \frac{16}{5}$.

Other Inequalities

Rearrangement (Permutation) Inequality: If $a_1 \geq a_2 \geq \dots \geq a_n$, and $b_1 \geq b_2 \geq \dots \geq b_n$,

then $a_1 b_1 + a_2 b_2 + \dots + a_n b_n \geq a_1 b_{r_1} + a_2 b_{r_2} + \dots + a_n b_{r_n} \geq a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1$, where

$r_1, r_2, \dots, r_n$ is a permutation of $1, 2, \dots, n$.

Chebyshev Inequality: If $a_1 \geq a_2 \geq \dots \geq a_n$, and $b_1 \geq b_2 \geq \dots \geq b_n$, then

$$\left(\frac{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}{n} \right) \geq \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) \left(\frac{b_1 + b_2 + \dots + b_n}{n} \right), \text{ with equality holds if and}$$

only if $a_1 = a_2 = \dots = a_n$ and $b_1 = b_2 = \dots = b_n$.

Power Mean Inequality: For $a_1, a_2, \dots, a_n > 0$, and $s < t$, then

$$M_s = \left(\frac{a_1^s + a_2^s + \dots + a_n^s}{n} \right)^{1/s} \leq M_t = \left(\frac{a_1^t + a_2^t + \dots + a_n^t}{n} \right)^{1/t}. \text{ Equality holds if and only if}$$

$$a_1 = a_2 = \dots = a_n.$$

Check that $M_1 = AM, M_{-1} = HM, M_0 = GM, \lim_{n \rightarrow \infty} M_n = \max(a_1, a_2, \dots, a_n)$, and

$\lim_{n \rightarrow -\infty} M_n = \min(a_1, a_2, \dots, a_n)$. M_2 is called the *root-mean-square* of the numbers.

Symmetric Mean Inequality: For $a_1, a_2, \dots, a_n > 0$,

$$AM = S_1 \geq S_2^{1/2} \geq \dots \geq S_n^{1/n} = GM, S_j \text{ is the average of all possible products of}$$

$a_1, a_2, \dots, a_n$ taken j at a time. Equality holds if and only if $a_1 = a_2 = \dots = a_n$.

Convex Functions

A continuous function is *convex on an interval I* if $f\left(\frac{x_1+x_2}{2}\right) \leq \frac{f(x_1)+f(x_2)}{2}$ for every $x_1, x_2 \in I$. The function is *strictly convex* if it is convex and the equality holds only when $x_1 = x_2$. A function $f(x)$ is *concave (strictly concave)* if $-f(x)$ is convex (strictly convex).

Second Derivative Test: If $f''(x) \geq 0$ on the open interval $I = (a, b)$, then $f(x)$ is convex on I , if $f''(x) > 0$, then it is strictly convex.

Examples of Strictly Convex Functions: x^p on $[0, \infty)$ for $p > 1$; x^p on $(0, \infty)$ for $p < 0$; a^x on $(-\infty, \infty)$ for $a > 1$; $\tan x$ on $[0, \frac{\pi}{2})$.

Examples of Strictly Concave Functions: x^p on $[0, \infty)$ for $0 < p < 1$; $\log_a x$ on $(0, \infty)$ for $a > 1$; $\cos x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$; for $a > 1$; $\sin x$ on $[0, \pi)$.

One of the most important inequality:

Jensen Inequality: $f(x)$ is convex on I and $x_1, x_2, \dots, x_n \in I$, then

$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \leq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}.$$

For strictly convex functions, equality holds if and only if all x_i are equal.

Generalised Jensen Inequality: $f(x)$ is convex on I and $x_1, x_2, \dots, x_n \in I$, $t_1, t_2, \dots, t_n > 0$ with $t_1 + t_2 + \dots + t_n = 1$, then

$$f(t_1 x_1 + t_2 x_2 + \dots + t_n x_n) \leq t_1 f(x_1) + t_2 f(x_2) + \dots + t_n f(x_n).$$

For strictly convex functions, equality holds if and only if all x_i are equal.

Example: Let $f(x) = \sin x$, it is strictly concave on $[0, \pi]$. Hence for ΔABC ,

$$\sin A + \sin B + \sin C = f(A) + f(B) + f(C) \leq 3f\left(\frac{A+B+C}{3}\right) = 3\sin\frac{A+B+C}{3} = \frac{3\sqrt{3}}{2}.$$

Equality holds if and only if $A = B = C = \frac{\pi}{3}$.

Example: If $a, b, c > 0$, and $a + b + c = 1$, find the minimum of

$$\left(a + \frac{1}{a}\right)^{10} + \left(b + \frac{1}{b}\right)^{10} + \left(c + \frac{1}{c}\right)^{10}.$$

In fact $0 < a, b, c < 1$, and $(x + \frac{1}{x})^{10}$ is strictly convex on $(0, 1)$, thus

$$\frac{10^{10}}{3^9} = 3f\left(\frac{1}{3}\right) = 3f\left(\frac{a+b+c}{3}\right) \leq f(a) + f(b) + f(c). \text{ The minimum is obtained when } a=b=c=\frac{1}{3}.$$

Example: Let $a_1, a_2, \dots, a_n > 0$, and as $f(x) = \log x$ is strictly concave on $(0, \infty)$, we have $\log\left(\frac{a_1 + a_2 + \dots + a_n}{n}\right) \geq \frac{\log a_1 + \log a_2 + \dots + \log a_n}{n} = \log \sqrt[n]{a_1 a_2 \dots a_n}$. Exponentiating to get $AM \geq GM$.

Apply Generalised Jensen Inequality to get $t_1 a_1 + t_2 a_2 + \dots + t_n a_n \geq a_1^{t_1} a_2^{t_2} \dots a_n^{t_n}$.

Example (Holder Inequality): $p, q > 1$, $\frac{1}{p} + \frac{1}{q} = 1$ and $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are real

numbers, then $\left| \sum_{i=1}^n a_i b_i \right| \leq \left(\sum_{i=1}^n |a_i|^p \right)^{1/p} \left(\sum_{i=1}^n |b_i|^q \right)^{1/q}$. (For $p = q = 2$, get the Cauchy-Schwarz Inequality.)

Let $A = \sum_{i=1}^n |a_i|^p$ and $B = \sum_{i=1}^n |b_i|^q$. If A or B is 0, then the inequality is valid. Otherwise, let

$t_1 = \frac{1}{p}$, $t_2 = \frac{1}{q}$, $x_i = \frac{|a_i|^p}{A}$ and $y_i = \frac{|b_i|^q}{B}$, then $\sum_{i=1}^n x_i = 1$ and $\sum_{i=1}^n y_i = 1$. Since $f(x) = e^x$ is strictly convex on $(-\infty, \infty)$, apply the generalised Jensen Inequality to get

$$x_i^{1/p} y_i^{1/q} = f(t_1 \ln x_i + t_2 \ln y_i) \leq t_1 f(\ln x_i) + t_2 f(\ln y_i) = \frac{x_i}{p} + \frac{y_i}{q}.$$

Sum $i = 1, 2, \dots, n$ and simplify to get the desired inequality.

Majorization Inequality: Suppose $x_1 \geq x_2 \geq \dots \geq x_n$, $y_1 \geq y_2 \geq \dots \geq y_n$,

$x_1 \geq y_1$, $x_1 + x_2 \geq y_1 + y_2$, ..., $x_1 + x_2 + \dots + x_{n-1} \geq y_1 + y_2 + \dots + y_{n-1}$, and

$x_1 + x_2 + \dots + x_n = y_1 + y_2 + \dots + y_n$. Then we say $(x_1, x_2, \dots, x_n)$ majorize $(y_1, y_2, \dots, y_n)$, written as $(x_1, x_2, \dots, x_n) \succ (y_1, y_2, \dots, y_n)$.

The inequality: $f(x)$ is convex and $(x_1, x_2, \dots, x_n) \succ (y_1, y_2, \dots, y_n)$, then

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

Example: ABC is an acute triangle, then $1 \leq \cos A + \cos B + \cos C \leq \frac{3}{2}$.

Assume $A \geq B \geq C$, then $\frac{\pi}{2} \geq A \geq \frac{\pi}{3}$, $C \leq \frac{\pi}{3}$, $\pi \geq A + B \geq \frac{2\pi}{3}$. We have $f(x) = \cos x$

strictly concave and $\left(\frac{\pi}{2}, \frac{\pi}{2}, 0\right) \succ (A, B, C) \succ \left(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}\right)$, by the majorization inequality

(reverse direction), get $1 = f\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) + f(0) \leq f(A) + f(B) + f(C)$

$$= \cos A + \cos B + \cos C \leq f\left(\frac{\pi}{3}\right) + f\left(\frac{\pi}{3}\right) + f\left(\frac{\pi}{3}\right) = \frac{3}{2}.$$